Zero-Days in Your Pocket: Comparing Android and iOS Under Real-World Exploitation

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***Abstract*—A zero-day vulnerability is exploited before any patch exists. On mobile devices, where the phone holds payment credentials, private messages, and location history, this class of flaw carries the highest risk. Android and iOS dominate the market, but they are built and shipped differently: Android is open source and reaches users through many manufacturers and update channels, while iOS runs on closed hardware controlled by a single vendor. This paper compares how the two platforms are affected by zero-days and how they respond to them. We compiled data from Google's in-the-wild tracker, vendor security bulletins, and the published literature, and compared Android and iOS along five dimensions: disclosure and patch delivery, ecosystem fragmentation, exploit mitigations, hardware-backed defenses, and documented in-the-wild exploitation. The synthesis finds that iOS accounts for most documented mobile zero-day exploitation, that Android's fragmented update chain slows patch delivery, and that both platforms now anchor their defenses in dedicated hardware. We close with implications for users, vendors, and future research.**

***Index Terms*—Android, iOS, hardware security, mobile security, vulnerability management, zero-day exploit**

I. Introduction

In January 2026, Google's Project Zero published a working zero-click exploit chain for a Pixel 9 phone [1]. An incoming audio message to Google Messages was decoded by a vulnerable Dolby decoder (CVE-2025-54957) before the user touched the device. The authors built the chain to study defensive options, and it shows what a well-funded attacker can do to a fully patched, current-generation device. Criminal capability on iOS is equally concrete: in 2025, the Citizen Lab forensically confirmed that Graphite, the spyware product of Paragon Solutions, had infected the iPhones of European journalists through a zero-click iMessage exploit (CVE-2025-43200) [2]. Neither case is unusual in kind. Google's Threat Intelligence Group (GTIG) tracked 97 zero-day vulnerabilities exploited in the wild in 2023, 75 in 2024, and 90 again in 2025 [3]–[5], and it observed that attackers chained several zero-days together almost exclusively against mobile devices [4]. The exposure is broad: Android runs on roughly 69% of smartphones worldwide [6], and Apple reported more than 2.5 billion active devices in January 2026 [7].

A zero-day is a flaw that is exploited before a patch exists, so no signature or prior update can stop the first attack [8]. How much damage that first attack does, and how fast it stops working, is decided by engineering choices made years earlier. Android is developed as open source, extended by device makers, and delivered to users through manufacturer-specific update channels. Google publishes a monthly security bulletin, but applying it is optional for each manufacturer [9]. iOS ships on hardware that Apple controls end to end, with one update pipeline and security fixes pushed directly to users, most recently as rapid security responses [10], [11]. These structural differences should be visible in the zero-day record: in how often each platform is exploited, how quickly fixes reach devices, and how far an attacker gets before hardware defenses stop them.

Both vendors have moved the anchors of that defense into dedicated hardware. Android pairs ARM TrustZone-based trusted execution environments with verified boot and hardware-backed keystores [12], [13]. Apple roots its boot chain in its own silicon and isolates secrets in the Secure Enclave [10]. A modern mobile exploit chain must therefore do more than corrupt memory: it has to bypass mechanisms built for exactly this purpose. A comparison of zero-day exposure that ignores this layer misses half the story.

Published work covers parts of this space. Surveys exist for Android malware and its defenses, for iOS exploitation techniques, and industry reports track zero-day counts year by year. What is missing is a structured, current comparison across both platforms that includes the hardware layer. This paper contributes a comparison framework with five dimensions for the zero-day posture of mobile operating systems; a synthesis of in-the-wild exploitation and patching data from 2019 through 2026, drawn from public trackers and vendor bulletins; and an evaluation that places hardware-backed defenses inside that comparison and derives implications for vendors and researchers.

The rest of the paper is organized as follows. Section II provides background on zero-day terminology and the security architectures of Android and iOS. Section III reviews related work. Section IV describes the comparison framework and the data collection. Section V evaluates the two platforms and discusses threats to validity. Section VI concludes.

II. Background

A. Zero-Day Fundamentals

A zero-day vulnerability is a flaw exploited before the vendor knows about it, so no patch exists during the first

attacks [8]. Its life cycle runs from private discovery through active exploitation to public disclosure, patch release, and eventual loss of value as the installed base updates; Fig. 1 sketches the timeline. Two properties make this cycle dangerous. First, exploitation is invisible for a long time: Bilge and Dumitras measured 18 vulnerabilities that were exploited in the wild for an average of 312 days before disclosure, with the longest case at about two and a half years [14]. Second, disclosure changes the threat rather than ending it; the same study found attack volume rose by up to five orders of magnitude after details became public [14].

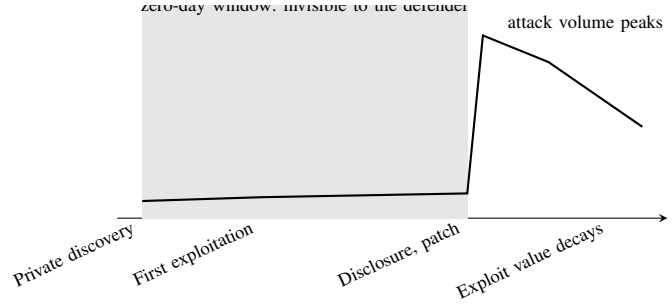


Fig. 1. Life cycle of a zero-day vulnerability. Attack volume is schematic, following the measurements of Bilge and Dumitras [14]: low and invisible during the shaded zero-day window, which averaged 312 days, peaking by up to five orders of magnitude after disclosure, then decaying as the installed base patches.

The RAND Corporation's study of more than 200 zero-days additionally found no evidence that zero-day exploits age quickly; many remained useful long after vendors patched them, because deployment of patches lagged [15]. Disclosure practice reflects this urgency: Project Zero gives vendors 90 days and cuts the deadline to seven days when a bug is found being exploited against real users [16]. Detection is the bottleneck of the whole field: Project Zero's tracker records, on average, a newly discovered in-the-wild exploit every 17 days [17], and each discovery means attackers were there first. Mobile platforms are a special case on both ends. Attacks are delivered without any user interaction, and Google's Threat Intelligence Group (GTIG) found that exploit chains combining several zero-days were aimed at mobile devices in about 90% of observed cases [4].

B. Android Security Architecture

Android is an open-source platform that Google does not ship to users directly. Device manufacturers adapt the Android Open Source Project (AOSP) to their hardware, and carriers sit between them and the subscriber. Google has published a monthly security bulletin since 2015 and requires a security patch level (SPL) with each release, but integrating the fixes into shipping devices is the manufacturers' work [9]. The measurable cost of this structure is delay. Jones et al. studied the deployment of Android security updates across manufacturers, carriers, and users, and found a median latency of 24 days before an update even begins to roll out, followed by further device-side delays [18]. Zhang et al. traced kernel security patches through the same ecosystem and documented the same pattern: OEM schedules introduce avoidable delay [19]. Google has narrowed the gap architecturally. Project Treble (Android 8.0, 2017) separated the OS framework from silicon-specific code, and modular system components, delivered as Google Play system updates since Android 10 (2019), let Google patch parts of the OS directly on every device [20]. Support contracts vary by manufacturer; Google's own Pixel 8 series carries a seven-year commitment [21]. The hardware layer holds the deepest anchors: verified boot checks each stage of startup against a hardware root of trust [12], ARM TrustZone isolates trusted execution environments (TEEs) from the main OS [22], and keys can reside in a separate secure processor via the hardware-backed keystore, such as the Titan M chip introduced with the Pixel 3 [13], [23].

C. iOS Security Architecture

iOS takes the opposite shape: Apple writes the OS, builds the SoC, reviews every App Store submission, and pushes updates to all supported devices through one channel [10]. The same guide documents the mitigation stack that has accumulated against zero-day exploitation. BlastDoor, added in iOS 14, parses untrusted Messages content inside a tightly sandboxed service that Project Zero's analysis credited with shrinking the iMessage attack surface [10], [24]. Lockdown Mode (iOS 16, 2022) is an opt-in mode for users facing mercenary spyware that disables the most exploitable surfaces [25]. Rapid Security Responses (iOS 16.4, 2023) deliver security fixes between full OS releases [11]. The hardware roots run deepest here: the immutable boot ROM anchors the boot chain, the Secure Enclave isolates biometric data and cryptographic keys on a dedicated coprocessor, ARM pointer authentication (A12, 2018) hardens code against memory-corruption reuse, and the Page Protection Layer (iOS 16) restricts modifications to kernel memory, with successors (SPTM, TXM) extending this from iOS 17 [10]. Every iPhone ships with this stack; nothing depends on a third party integrating it.

III. Related Work

Research relevant to this paper falls into three groups.

A. Empirical Zero-Day Studies

Bilge and Dumitras pioneered the measurement of zero-day attacks in the wild and quantified how long exploitation precedes discovery [14]. Ablon and Bogart described the economic side: who buys zero-days, what they pay, and how long exploits stay useful [15]. Since 2019, Google has published the definitive exploitation corpus. Project Zero tracks every publicly known in-the-wild zero-day in a public list [17], and GTIG's annual reviews break the counts down by vendor, product type, and attacker [3]–[5]. These sources describe what happened but stop short of comparing platforms systematically.

B. Platform Security Comparisons

Faruki et al. surveyed Android's security mechanisms and malware landscape [26]. The closest work to ours is the comparison by Garg and Baliyan, who contrasted Android and iOS across architecture, app stores, permission models, and updates, and concluded that Android's fragmented update ecosystem makes it the more vulnerable platform [27]. Their study covers the state of both systems through 2020. Several of the mechanisms this paper evaluates (Lockdown Mode, Page Protection Layer, mature Google Play system updates) postdate their analysis, and their data set contains no in-the-wild exploitation counts.

C. Hardware-Backed Security and Spyware Cases

Pinto and Santos surveyed TrustZone-based TEEs [22], and Cerdeira et al. systematized the vulnerabilities found in deployed TEEs [28]. Forensic reports from the Citizen Lab document individual iOS zero-day campaigns in technical depth, among them FORCEDENTRY [29] and the Paragon campaigns [2], and Amnesty International's methodology report established the indicators used to detect Pegasus infections at scale [30]. Google Project Zero's own writing covers both sides: the discovery of BlastDoor [24] and a full zero-click Android research chain [1].

The present work differs from all of the above in three ways. It compares the two platforms on zero-days specifically rather than on security in general; it draws on exploitation data from 2023 through 2026, after the latest hardening rounds on both sides; and it places hardware-backed defenses inside the comparison rather than treating them as a separate literature.

IV. Implementation: Comparative Framework

A. Method

The study is a literature meta-analysis. We collected sources in four categories: annual in-the-wild exploitation reviews from GTIG [3]–[5] and the Project Zero tracker [17]; forensic case reports on mobile spyware campaigns [2], [29], [30]; official vendor documentation for security bulletins, platform security, and update mechanisms [9], [10]; and peer-reviewed empirical studies of mobile security and patching [14], [18], [19], [27]. The window of interest is 2019 through 2026, starting when continuous in-the-wild tracking became available. A source was included if it documents a confirmed in-the-wild mobile zero-day, a measured property of the disclosure or patching pipeline, or a deployed defense mechanism on either platform. From each included source we extracted, where applicable: the CVE identifier, platform, delivery vector, fix version and date, measured patch latency, and the hardware or software mechanism the case involved. The extraction produced the data behind Section V. The final corpus comprises three annual exploitation reviews, the continuous Project Zero tracker, three forensic case reports, seven peer-reviewed studies, and primary vendor documentation and statistics, and the per-case extraction is archived in machine-readable form alongside the collected sources.

A meta-analysis is the appropriate instrument here because the phenomenon cannot be measured directly: re-running attacks against stock devices would reproduce only the research side of the picture, while the criminal side exists solely in detections and forensic reports. The method trades experimental control for breadth, and its weaknesses are treated explicitly in Section V-D.

B. Framework

The comparison framework has five dimensions, each tied to an observable indicator. Table I defines them and lists what is compared on each platform.

TABLE I
Comparison Framework: Dimensions and Indicators

Dimension	Android indicator	iOS indicator
Disclosure and patch delivery	Monthly bulletin, SPL schedule, Play system updates	Direct OTA to all devices, Rapid Security Responses
Ecosystem fragmentation	OEM and carrier latency (days)	Single vendor, one channel
Exploit mitigations	SELinux, seccomp, app sandbox	BlastDoor, Lockdown Mode, PAC
Hardware-backed defenses	Verified boot, TrustZone TEE, Titan M keystore	Boot ROM, Secure Enclave, PPL
In-the-wild exploitation	ITW zero-day counts, CSV share	ITW zero-day counts, CSV share

Dimensions one to three capture how fast a fix becomes protection and how hard exploitation is once it starts. Dimension four captures the seminar's theme: the anchors that survive even a fully compromised OS kernel. Dimension five is the outcome measure, taken from the GTIG corpus, that shows how the structural differences above translate into recorded attacks. Applying an outcome measure alongside structural measures is what distinguishes this framework from a feature checklist.

Threat model. We consider four actor classes, taken from the attribution in the sources: state-sponsored espionage groups, which GTIG consistently ranks among the most prolific zero-day users [3]; commercial surveillance vendors (CSVs), which sell exploit chains to state customers [2], [29]; physical-access forensic toolmakers such as Cellebrite [4]; and financially motivated criminals, largely absent from the mobile record because their zero-day targets lie in enterprise software [5]. The attacker starts remotely with no user interaction or, in the forensic class, with the device in hand. Success means arbitrary code execution, and full compromise requires defeating the hardware-backed boundary. The framework scores defenses against exactly this progression.

V. Evaluation

A. In-the-Wild Exploitation

Table II summarizes the GTIG counts, and Fig. 2 shows how small the mobile share of total exploitation is. Mobile zero-days fell from 17 in 2023 to 9 in 2024 and rose again

to 15 in 2025 [3]–[5]. The counts understate mobile's role: in 2024, exploit chains that combined several zero-days targeted mobile devices in roughly 90% of observed cases [4].

TABLE II
ZERO-DAYS EXPLOITED IN THE WILD, AS FIRST REPORTED BY GTIG

Year	Total ITW	Mobile	Android	Apple
2023	97	17	7	–
2024	75	9	7	5
2025	90	15	–	–

Dashes: per-vendor counts not itemized in that year's review. Later reviews revise past totals upward (2023 to 100, 2024 to 78) as exploitation is back-attributed [5], so every figure is a lower bound.

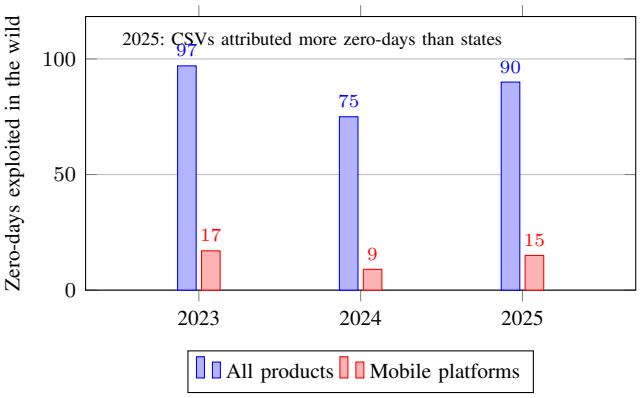


Fig. 2. In-the-wild zero-day counts per year. Mobile is a minority of exploitation but carries almost all multi-zero-day chains.

Who exploits, and what, differs by platform. Commercial surveillance vendors (CSVs) dominate this space: they were behind 75% (13 of 17) of the zero-days that hit Google products and Android ecosystem devices in 2023, and more than 60% of the 37 browser and mobile zero-days that year [3]. In 2025, GTIG attributed more zero-days to CSVs than to state-sponsored espionage groups for the first time [5]. The 2025 mobile rebound has identifiable causes: researchers uncovered more complete chains, chains grew to three or more vulnerabilities, and vendors such as Intellexa developed new techniques to cross the security boundaries Apple had added [5]. Browsers show what successful platform hardening looks like: they drew under 10% of 2025 zero-days, a historic low that GTIG credits to years of sandboxing and memory-safety work [5]. Mobile OS hardening has not produced a comparable drop, and the case evidence below suggests why. On Android, the exploited bugs concentrate in third-party components, which scale one exploit across many device makers: five of seven Android zero-days in 2023 and three of seven in 2024 [4].

Table III lists the four anchor cases, and Fig. 3 places their verified dates on one axis. The iOS campaigns share a recipe, zero-click delivery followed by an emergency fix, and FORCEDENTRY bypassed BlastDoor, the very sandbox added to stop it [29]. The Android cases bookend the spectrum: a zero-click research chain through a third-party decoder [1] and a forensic tool needing physical access [4].

TABLE III
DOCUMENTED IN-THE-WILD AND RESEARCH CASES

Case (actor)	Platform	Vector	Fix
FORCEDENTRY (NSO, 2021)	iOS	Zero-click iMessage, CVE-2021-30860	iOS 14.8
Graphite (Paragon, 2025)	iOS	Zero-click iMessage, CVE-2025-43200	iOS 18.3.1
Cellebrite chain (2024)	Android	Physical USB access, CVE-2024-53104	Bul. Feb. 2025
Pixel 9 chain (P0, 2026)	Android	Zero-click transcription, CVE-2025-54957	Before disclosure

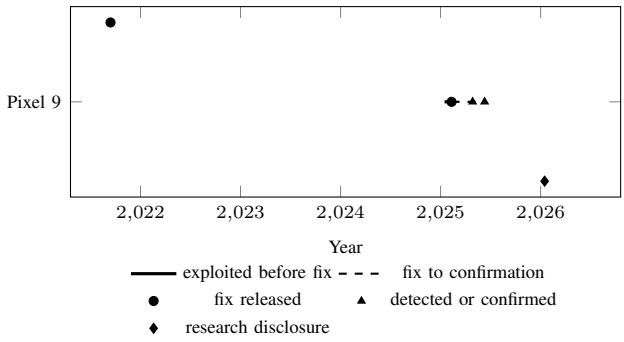


Fig. 3. Case timeline assembled from Table III. FORCEDENTRY was captured and fixed on the same day, so its pre-detection span is unknown. Graphite infections ran about a month before the iOS 18.3.1 fix, and forensic confirmation followed four months later.

B. Applying the Framework

Table IV fills the five dimensions of Table I with the evidence collected above.

The synthesis runs against the older narrative that Android is simply the less secure platform [27]. On delivery speed and reach, iOS holds the advantage: a fix Apple ships reaches every supported device, while the same Android fix waits on OEM integration, measured in weeks before rollout even starts [18]. On attack surface, the platforms fail differently. Android's exploited zero-days sit mostly in third-party components, a structural consequence of the ecosystem's openness; iOS draws sustained, sophisticated zero-click campaigns from the spyware market, and its mitigation stack is exactly what those campaigns must defeat. On hardware, both platforms anchor their defenses in silicon, and the difference is universality: every iPhone carries the boot ROM, Secure Enclave, and PPL, while Android's deepest anchors (a discrete secure processor) depend on the device maker. The Pixel 9 chain adds a third-party illustration of the same point: the vulnerable Dolby decoder code also ships on Apple platforms, where the same bug class was blunted in advance by the `-fbounds-safety`

TABLE IV
FRAMEWORK APPLIED: ANDROID AND IOS COMPARED ON THE FIVE DIMENSIONS

Dimension	Android	iOS
Patch delivery	Monthly bulletin; OEMs integrate fixes; median 24-day rollout latency before updates even begin [18]; Play system updates patch some components directly [20]	One vendor, one channel; Rapid Security Responses push fixes between releases [11]
Fragmentation	Hundreds of manufacturer and carrier variants; kernel patches propagate unevenly [19]; Pixel 8 sets a 7-year ceiling [21]	Single hardware and OS vendor; support horizon uniform across devices [10]
Exploit mitigations	SELinux, seccomp, app sandbox; no platform-wide messaging sandbox; the Pixel 9 chain entered through a third-party decoder [1]	BlastDoor isolates message parsing [24]; Lockdown Mode for high-risk users [25]; FORCEDENTRY shows bypasses are possible [29]
Hardware defenses	Verified boot and TrustZone broadly present [12], [22]; secure processors (Titan M) ship on some devices [23]; deployed TEEs have documented vulnerabilities [28]	Boot ROM, Secure Enclave, PAC, and PPL present on every device [10]
In-the-wild outcome	7 ITW zero-days in each of 2023 and 2024, mostly third-party components [4]; physical-access forensic chains [4]	Repeated zero-click CSV campaigns (2021, 2025) [2], [29]; 5 ITW zero-days in 2024 [4]

compiler mitigation [1]. Garg and Baliyan's conclusion still holds for updates; it does not hold for in-the-wild exploitation, where the more integrated platform is the one spyware vendors keep paying for.

C. Implications

For users, the timing of updates matters in a specific way. The first attack is largely unavoidable, because exploitation is invisible for months, but the wave after disclosure is not: attack volume grew by up to five orders of magnitude once details became public in the measurement of Bilge and Dumitras [14], so updating promptly removes most of the real-world risk. Users facing targeted spyware have one further option on iOS that Android lacks platform-wide: Lockdown Mode trades functionality for a smaller attack surface [25]. For vendors, the evidence points to two levers. Shortening the path from fix to installed base is the first, where Apple's direct channel and Android's modular system updates both matter [11], [20]. Governing third-party components is the second, since they supplied most of the exploited Android bugs. For researchers and regulators, the binding constraint is detection: every count in this paper is an exploitation that someone happened to notice [17], and update-focused regulation can only work through the patching levers already discussed.

D. Threats to Validity

In-the-wild counts are detections, not attacks: an exploit used quietly against few targets can stay invisible, and mobile forensics skew toward high-value victims, which may inflate iOS's visibility relative to Android. Attribution to actor classes (CSV, state-sponsored) carries uncertainty in the underlying reports. GTIG revises past-year counts upward as new evidence arrives, so all totals are lower bounds. A sensitivity check under the latest attribution (100 for 2023, 78 for 2024) changes no conclusion: mobile remains a small minority of exploitation, the per-vendor ordering stands, and the CSV milestone is a 2025 event under either base. Vendor transparency differs; we compare the two platforms with the best public reporting, which limits generalization to other mobile systems. Finally, cross-platform messaging apps blur the boundary: the Paragon campaign used a WhatsApp flaw alongside the iOS exploit [2], and such bugs are not cleanly attributable to either OS.

VI. CONCLUSION AND FUTURE WORK

This paper compared Android and iOS through a five-dimension framework covering disclosure and patch delivery, fragmentation, exploit mitigations, hardware-backed defenses, and documented in-the-wild exploitation. Three findings stand out. First, the platforms fail differently: Android's zero-day exposure is dominated by fragmented patch delivery and third-party components, while iOS attracts repeated zero-click chains from commercial surveillance vendors. Second, patch speed differs structurally: iOS delivers fixes directly to every supported device, whereas Android updates wait on manufacturer integration with a measured median latency of 24 days. Third, both platforms now anchor their defenses in dedicated hardware, differing mainly in whether every device ships the full stack. The evidence base is detection-limited, so the counts should be read as lower bounds. Future work should track the effect of memory tagging on exploit costs as it rolls out, examine regulation such as the EU Cyber Resilience Act as an update-forcing mechanism for the Android ecosystem, and extend the framework to additional mobile operating systems.

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